IEOR 4101: Probability, Statistics and Simulation

Lecture 4: Bayes' Formula

Bayes' Formula

- Often applied in situations where there is a question of a hypothesis being true or false
  - E.g., whether a defendant is guilty; whether someone has a disease

- $H = \{\text{hypothesis is true}\}$, $H^c = \{\text{it is false}\}$

- Before examining evidence, *prior* probabilities $P(H)$ and $P(H^c)$
  - E.g., ≈ 0.2% of HIV carriers; $P(H) \approx 0.2\%$

- How do the probabilities change after the evidence E?

Bayes' Formula

- Interested in P(H|E) – *posterior* probability of H given E
- Suppose we have known P(E|H) and P(E|H^c) on top of P(H), then we can derive P(H|E) as follows

$$
\begin{aligned}
P(H \mid E) &= \frac{P(H \cap E)}{P(E)} \\
&= \frac{P(E \mid H)P(H)}{P(E)} \\
&= \frac{P(E \mid H)P(H)}{P(E \mid H)P(H) + P(E \mid H^c)P(H^c)}
\end{aligned}
$$

- Second equality follows from multiplication rule
- Third equality follows from the law of total probability

Bayes' Formula

- **Interpretation**:
  - P(H) reflects our *a priori* belief about hypothesis H
  - P(H|E) reflects our *a posterior* belief about H, given the new evidence E

Calculating $P(H|E)$ needs
– $P(E|H)$ and $P(E|H^c)$, and
– the prior probabilities $P(H)$ (or $P(H^c)$)

Bayes' Formula

- **Example** (Rare disease problem). Suppose we have a blood test which is 99% effective in detecting a certain disease when it is present. The test also yields a "false positive" for 2% of the healthy population tested. If 0.1% of the population has the disease, what is the probability that a person has the disease given that his result is positive?

H = disease

E = tested positive

$P(H) = 0.1\%$ prior probability

$P(H|E)$ posterior probability

$$P(H|E)$$

$$= \frac{P(H)P(E|H)}{P(H)P(E|H) + P(H^c)P(E|H^c)}$$

$$= \frac{0.1\% \times 99\%}{0.1\% \times 99\% + 0.99\% \times 2\%}$$

Bayes' Formula

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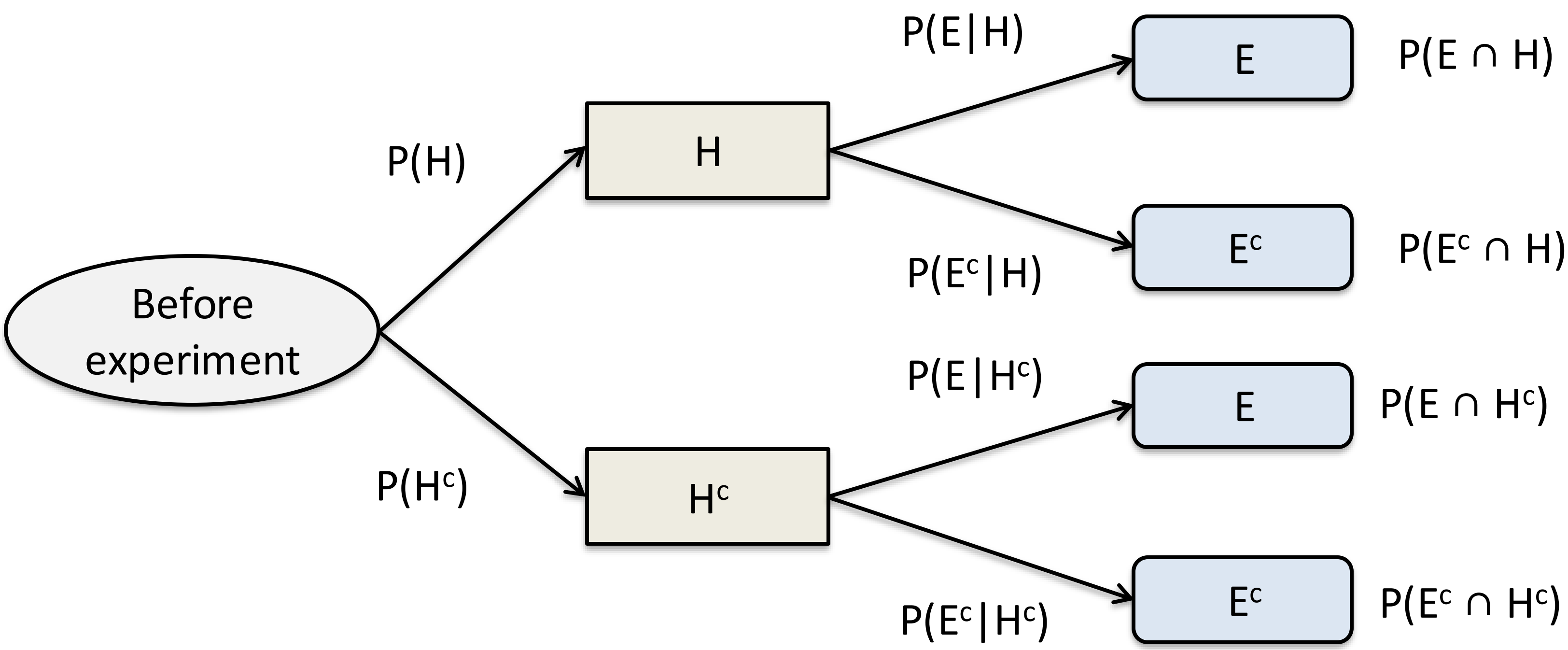
- **Setup and Solution**.
  - D = {disease}; +ve = {test result is positive}.
  - Know that P(+ve|D) = 0.99, P(+ve|D^c) = 0.02, P(D) = 0.001
  - $$P(D|+ve) = \frac{P(+ve|D)P(D)}{P(+ve|D)P(D) + P(+ve|D^C)P(D^C)}$$

$$= \frac{0.99 \times 0.001}{0.99 \times 0.001 + 0.02 \times 0.999} = 0.0472$$

The final answer 4.72% looks low, but it's not when compared to prior 0.1%.

Bayes' Formula

- Can also use chance tree



Bayes' Formula

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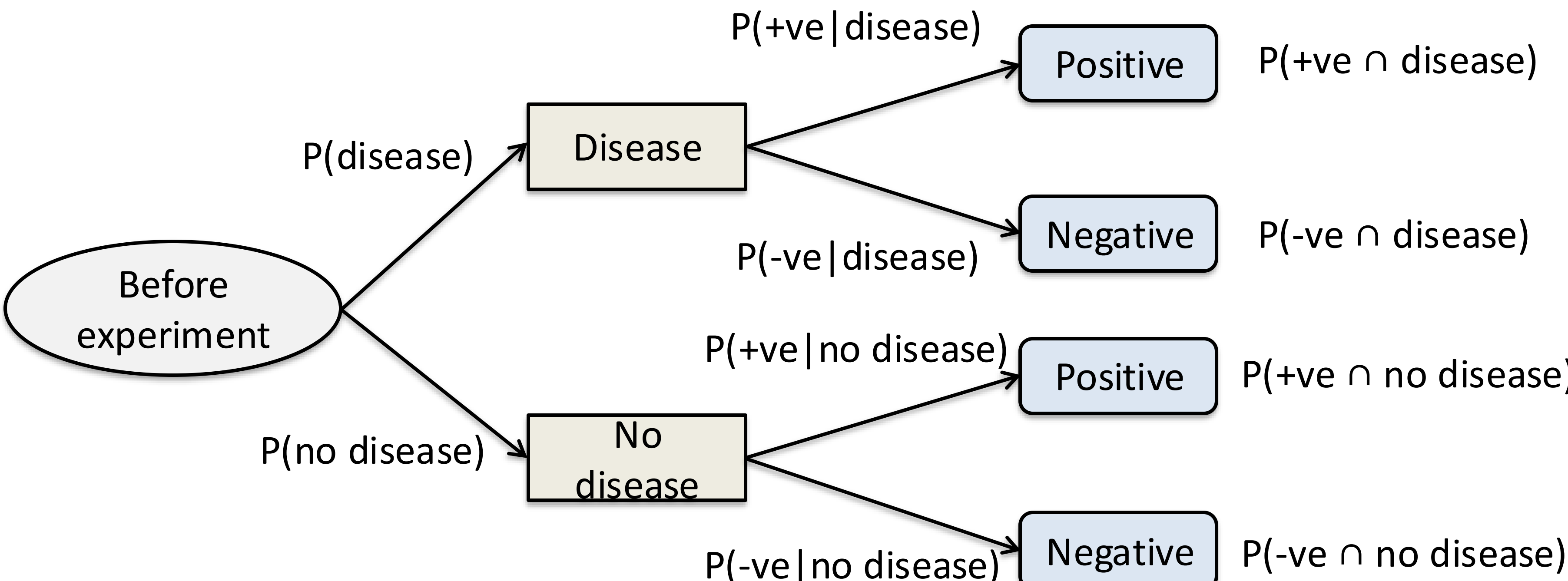
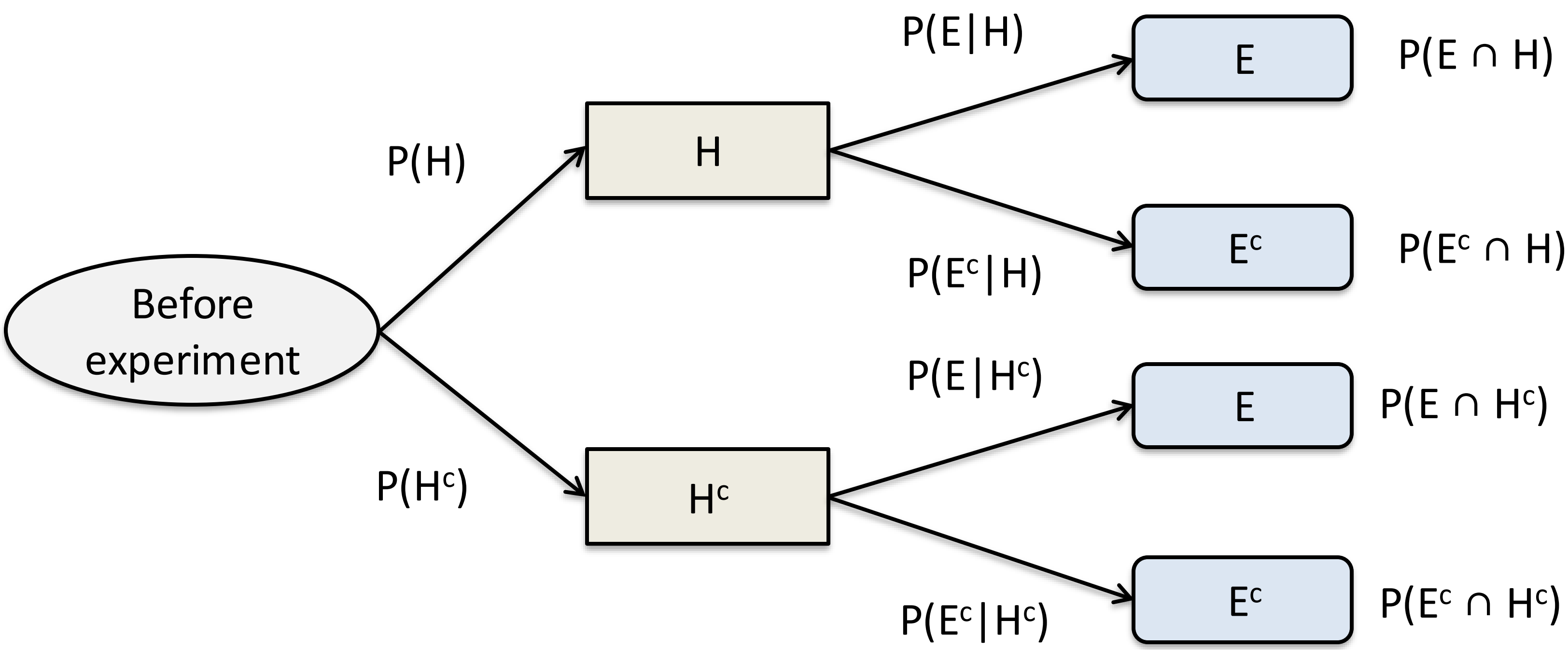


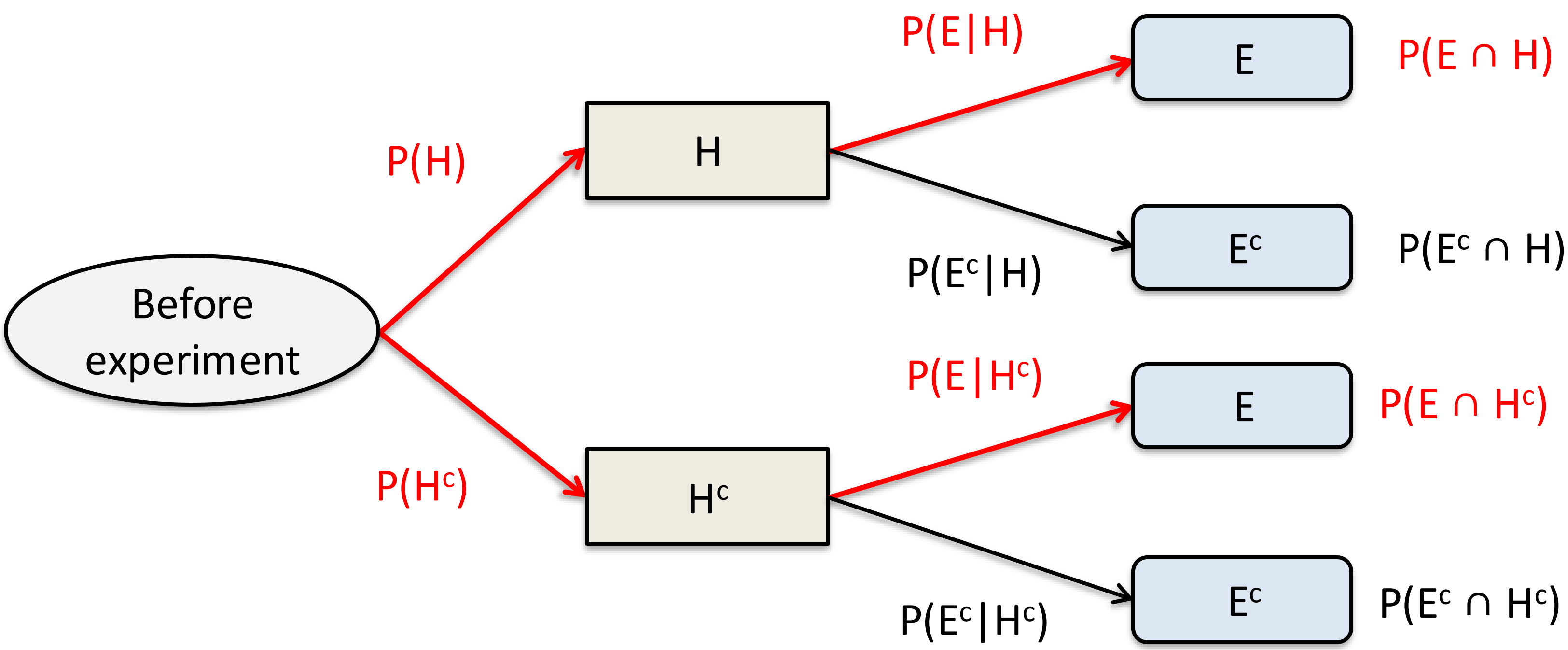
Bayes' Formula

- To calculate $P(H|E)$,



Bayes' Formula

- To calculate $P(H|E)$,
- E.g., $P(E \cap H) = P(E|H)P(H)$
- $P(E) = P(E \cap H) + P(E \cap H^c) = P(E|H)P(H) + P(E|H^c)P(H^c)$
- Ratio gives $P(H|E)$



Bayes' Formula

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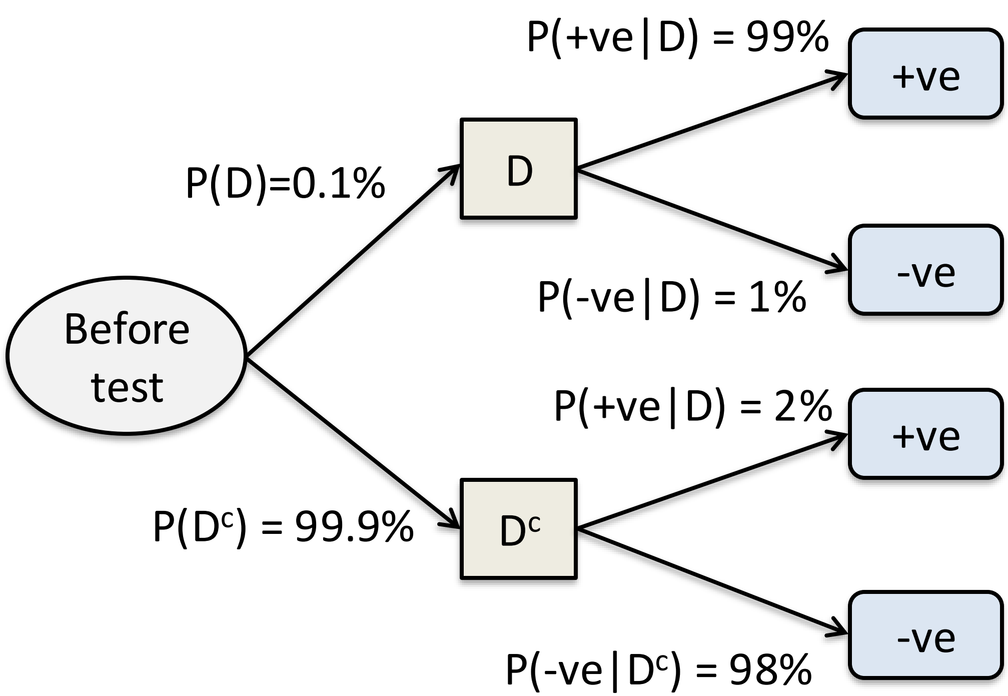

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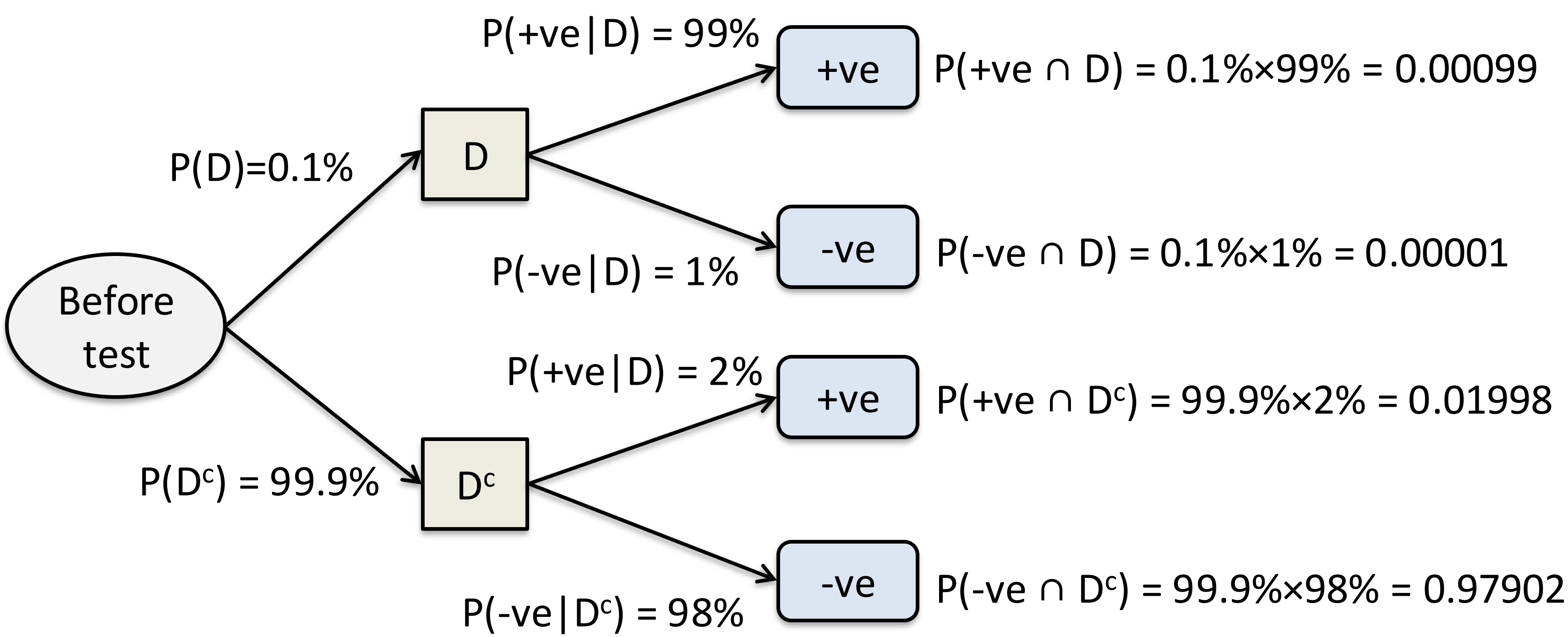

$$P(D \mid +ve) = \frac{0.00099}{0.00099 + 0.01998}$$

Bayes' Formula

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- **Setup and Solution**.
  - D = {disease}; +ve = {test result is positive}.
  - Know that $P(+ve|D) = 0.99$, $P(+ve|D^c) = 0.02$, $P(D) = 0.001$
  - From the chance tree:
    - $P(+ve \cap D) = 0.00099$, $P(+ve \cap D^c) = 0.01998$
    - $P(+ve) = P(+ve \cap D)+P(+ve \cap D^c) = 0.00099+0.01998 = 0.20079$
    - $P(D|+ve) = P(+ve \cap D) /P(+ve) = 0.00099/0.20079 = 0.0472$